

## INTER QUARTILE RANGE

IQR (Interquartile Range) is the difference between the third quartile (Q3) and the first quartile (Q1).

Formula:

$$\text{IQR} = Q3 - Q1$$

It represents the spread of the middle 50% of the data.

In Data Science, we use IQR especially to **find outliers**.

$$\text{LESSER QUARTILE} = Q1 - 1.5 \times \text{IQR}$$

$$\text{GREATER QUARTILE} = Q3 + 1.5 \times \text{IQR}$$

**Why use 1.5 rule?**

Because **1.5 is a standard practical rule** for finding potential outliers.

**$1 \times \text{IQR}$**  → boundary is too close → may call many normal values outliers.

**$1.5 \times \text{IQR}$**  → gives a reasonable extra distance → commonly used.

**$3 \times \text{IQR}$**  → boundary is very far → may miss some unusual values.

Example:

- $Q1 = 20$
- $Q3 = 60$
- $IQR = 40$

**Using  $1 \times IQR$ :**

$$\text{Upper limit} = 60 + 40 = \mathbf{100}$$

**Using  $1.5 \times IQR$ :**

$$\text{Upper limit} = 60 + 60 = \mathbf{120}$$

**Using  $3 \times IQR$ :**

$$\text{Upper limit} = 60 + 120 = \mathbf{180}$$

1.5 is used because it is a standard rule of thumb that provides a practical boundary for detecting potential outliers. 1 may be too sensitive, while 3 may be too relaxed.”